Course Syllabus

Course Information

CS/CE 4337.006 Programming Language Paradigms - Spring 2026
Lecture Info: TuTh: 8:30 am - 9:45 am ; Class Room location: JO 4.614

Professor's Contact Information

Professor: Dr. Gity Karami
Office Phone:972-883-4204
Office Location:ECSS 3.202
Email: gity.karami@utdallas.edu
Office hours: Tu: 12 pm – 1 pm by appointment
Th: 12 pm – 1 pm Walk-in
Signup link: https://calendly.com/gxk180009/office-hours
Please sign up in advance (at least one day before your scheduled meeting)

Course Modality and Expectations

- **Instructional Mode:** Traditional Classroom/Laboratory
- **Course Platform:** All instruction will be through the eLearning platform, where all the course materials including links to the recorded/online resources will be posted. We will also use Microsoft Team for Virtual office hours and piazza as asynchronous Q&A platform.
- **Expectations:** Regular and punctual class attendance is expected. Students who fail to attend class regularly are inviting scholastic difficulty. Active participation on piazza is also expected.

Class Recordings:

The class's lectures may be recorded and the links posted in eLearning. Additionally, the instructor may record other meetings of this course, and such recordings will be posted in eLearning too. **Students are expected to follow appropriate University policies and maintain the security of passwords used to access recorded material**. Unless the Office of Student AccessAbility has approved the student to record the instruction, students are expressly prohibited from recording any part of this course. **Recordings may not be published, reproduced, or shared with those not in the class, or uploaded to other online environments** except to implement an approved Office of Student AccessAbility accommodation. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Piazza:.

We'll be using piazza as asynchronous Q&A platform. The quicker you begin asking questions on Piazza (rather than via emails), the quicker you'll benefit from the collective knowledge of your classmates and instructor. I encourage you to ask questions when you're struggling to understand a concept. The link to enroll in Piazza is available in e-learning.

Microsoft Team:

We'll be using Microsoft Team for Virtual office hours. Virtual office hours will be held using Microsoft Team on Tu: 12 pm – 1 pm by appointment; **Signup link:** https://calendly.com/gxk180009/office-hours
Please sign up in advance (at least one day before your scheduled meeting). Live lectures may also be available on Microsoft Team during the class time.

Class Participation:

Regular and punctual class attendance is expected. Students who fail to follow the class materials regularly are inviting scholastic difficulty. The course's material gets much more complex as the course progresses, and it is typically very difficult to catch up with missed classes. Active participation on piazza is also expected.

Class Materials:

The instructor may provide class materials that will be made available to all students registered for this class. These materials may be downloaded during the course, however, these materials are for registered students' use only. Classroom materials may not be reproduced or shared with those not in the class or uploaded to other online environments, except to implement an approved Office of Student AccessAbility accommodation. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Course Prerequisites:

((CE 2336 or CS 2336 or CS 2337) with a grade of C or better or CS 3333) and (CE 2305 or CS 2305) with a grade of C or better and (CS 2340 or SE 2340 or CE 4304 or EE 4304). (Same as CS 4337) (3-0) S.

Course Description:

CS/CE 4337-**Programming Language Paradigms** (3 semester credit hours) Principles of design and implementation of contemporary programming languages. Formal description including specification of syntax and semantics of programming languages. Language definition structures including binding, scoping, data types, control structures, parameter passing, abstraction mechanism, and run-time considerations. Design issues of different programming languages. Language-based security. Design, implement, and debug programs in various programming language paradigms.

Text Book:

Concepts of Programming Languages,12th Edition, Robert W. Sebesta. Pearson, 2019, ISBN-10: 0-13-499718-2, ISBN-13: 978-0-13-499718-
*supplementary materials (Research papers and handout) will be posted in the e-learning

Learning Objectives:

- Ability to identify the characteristics of programming paradigms and phases of translation
- Ability to understand the concepts of formal syntax and semantics of programming languages
- Ability to understand the concept of language-based security
- Ability to understand the concept of web scripting security
- Ability to understand the concepts of first-class values, lists, and recursion
- Ability to understand the concepts of the functional programming paradigm and design functional programs
- Ability to understand the concepts of the logic programming paradigm and design logic programs

Tentative Course Schedule:

Week	Date	Material Covered
1	Jan 20, Jan 22	Syllabus, characteristics of programming paradigms (Sebesta-Ch1), Functional programming paradigm (Sebesta- Ch15)
2	Jan 27, Jan 29	Functional programming paradigm (Sebesta-Ch15)
3	Feb 3, Feb 5	Functional programming paradigm (Sebesta-Ch15)
4	Feb 10, Feb 12	Describing syntax and semantic (Sebesta-Ch3)
5	Feb 17, Feb 19	Describing syntax and semantic (Sebesta-Ch3)
6	Feb 24, Feb 26	Lexical and syntax analysis (Sebesta-Ch4)
7	Mar 3, Mar 5	Lexical and syntax analysis (Sebesta-Ch4)
8	Mar 10, Mar 12	Common Weakness Enumeration (CWE), Language based security (type-safe vs. type-unsafe languages)
9	Mar 17, Mar 19	Spring Break
10	Mar 24, Mar 26	Language based security (type-safe vs. type-unsafe languages), Midterm Exam
11	March 31, Apr 2	Logical programming paradigm (Sebesta-Ch16)
12	Apr 7, Apr 9	Logical programming paradigm (Sebesta-Ch16)
13	Apr 14, Apr 16	Language based security (layered security) and Web Scripting Security
14	Apr 21, Apr 23	Language based security (layered security) and Web Scripting Security, Language based security (Formal method)
15	Apr 28, Apr 30	Language based security (Fuzzing)
16	May 5, May 7	Final Exam , Language based security (control-flow integrity)

Course Works and Grading Policies:

Exams: 60% (Quizzes: 5 %, Midterm Exam: 20% , Final Exam: 35%)
Assignments: 40% (Four Assignments, each 10%)

*Instructor reserves the right to alter these weights or make changes as she sees fit.

Grades will be assigned according to the following scale

A+	97 and above
A	94 - 96 (94 or more and less than 97)
A-	90 - 93 (90 or more and less than 94)
B+	87 - 89 (87 or more and less than 90)
B	84 - 86 (84 or more and less than 87)
B-	80 - 83 (80 or more and less than 84)
C+	77 - 79 (77 or more and less than 80)
C	74 - 76 (74 or more and less than 77)
C-	70 - 73 (70 or more and less than 74)
D	60 - 69 (60 or more and less than 70)
F	Below 60

Exam: This course consists of two exams: a midterm exam (20% of the final grade) and a comprehensive final exam (35% of the final grade). Both exam grades will be factored into your final grade. No exam grades will be dropped. You are responsible for being available during the exam times. If you cannot make an exam time due to a valid excuse, you must let me know BEFORE the exam date and time. Medical emergencies will require a note from your Doctor. Missed exam will result in a grade of 0 for that exam.

Assignments: Doing assignments is vital for meeting the learning objectives and succeeding in this course. There will be four assignments in this course. You must work on the assignments with a partner and submit them on e-learning. No e-mail submissions are accepted. No late submissions are accepted. So, please plan accordingly, do not leave your submissions to the last minute. Everybody submits his/her work very easily via e-Learning, you can do it, too. If you encounter a problem during e-Learning submission, please contact 24/7 e-Learning Help IMMEDIATELY. This help is available 24/7 at:

e-Learning Help URL: http://www.utdallas.edu/elearning/eLearningHelpdesk.html
e-Learning Help Phone: 1 866 588 3192

If you believe that you have a valid excuse for your work being late, then you must make arrangements with the instructor BEFORE the due date. Late submissions are not permitted once the graded assignment has been returned to students. Medical excuses will require a note from your Doctor.

Quizzes: You are supposed to work on each quiz individually or with a partner. You are allowed to use the text book and lecture slides during the quizzes. You must submit them on e-learning. No e-mail submissions are accepted.

Grading Disputes: All grade disputes must be submitted in person via dispute form within 3 days of the grade being posted in eLearning. Assignments (graded

by the TA/Grader) should be disputed with the TA/Grader, while exams and quizzes (graded by the Instructor) should be disputed with the Instructor. Uncontested grades will become final after 3 days and cannot be disputed later.

Academic Dishonesty: You should do your own work on exam and assignments. Copying another student's work is not acceptable. Any indication of cheating and/or plagiarism on an exam/assignment will be an automatic 0 (zero) for the exam/assignment for all students involved. Solutions copied from the internet, instructor's manual, etc. will be also given zero credit. Please note that suspected incidents will be reported to the Office of Community Standards and Conduct.

Communications: I will be communicating with you via eLearning, piazza, and e-mail. If you need to send me an e-mail make sure it is using your UTD e-mail address. Please choose appropriate subjects for your emails. Always include your course and section number in the subject of your emails (for example, CS 4337.001- Exam). I won't answer your emails, if you don't put the course number and section number in the subject of your emails. Per FERPA guidelines, grade discussions are not allowed via email. Make sure you are checking grades and announcements posted in e-learning and checking your UTD e-mail frequently. I can't respond to you via gmail or any other non-UTD e-mail system. I need to verify that you are my student and I can only do that with the UTD e-mail system.

Comet Creed: "As a Comet, I pledge honesty, integrity, and service in all that I do."

Additional Policies: Please visit http://go.utdallas.edu/syllabus-policies for all other University policies

Descriptions and timelines are subject to change at the discretion of the Professor.